NLP-Based Intelligent Web Application Development

Project Title: NLP Assistant - Intelligent Chatbot and Text Analyzer

Overview:
This Flask web application processes human language through typed text or browser voice input.
It returns chatbot responses, visible NLP analysis, speech output, and database-backed history.

Technology Stack:
- Selected Frontend: HTML
- Supporting Frontend Assets: CSS, Bootstrap, and JavaScript for styling and browser interactivity
- Backend: Python Flask
- NLP: NLTK with rule-based fallbacks
- Database: SQLite runtime database and SQL export file
- Speech: Browser Web Speech API and Speech Synthesis API

Implemented NLP Concepts:
1. Tokenization - displays tokenized user input.
2. Sentiment Analysis - classifies Positive, Negative, or Neutral.
3. Voice-to-Text - microphone input converts speech to text.
4. Text-to-Speech - Speak button reads chatbot output aloud.
5. Chatbot System - user vs bot interface with stored history.
6. Keyword Extraction - highlights important words from input.
7. Text Classification - labels messages as Complaint, Inquiry, Feedback, Command, or General.

System Workflow:
User input -> JavaScript fetch request -> Flask route -> NLP engine -> database save -> JSON response -> UI displ

Database Table: chats
Fields: id, user_message, bot_response, tokens, keywords, sentiment, sentiment_score, classification, created_at.

How to Run:
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python3 app.py
Open http://127.0.0.1:5000

Defense Demo Checklist:
Show chatbot, token list, sentiment result, keywords, classification, voice input, speech output, and history table.